Skills Summary

Sigma Notation: Index Shifts and Convergence

Use this reference while completing your worksheet or when studying.

Skill 1 — Shifting the index of a finite sum

Rule: a sum is its summand *and* its limits. Substituting $m = n - c$ moves **every** index value, so both limits move by the same c:

$$\sum_{n=p}^{q} f(n-c) \;=\; \sum_{m=p-c}^{q-c} f(m)$$

Term count (fencepost): the sum $\sum_{n=p}^{q}$ has $q - p + 1$ terms. Counting terms before and after a shift is the fastest way to catch a bad one.

Example: Rewrite $\sum_{n=2}^{5}(n-1)^2$ with the index starting at 1, then evaluate.
Step 1. The summand is a function of $n - 1$, so substituting $m = n - 1$ simplifies it — and that same substitution must be applied to both limits.
Step 2. $n = 2$ gives $m = 1$ and $n = 5$ gives $m = 4$, so the sum is $\sum_{m=1}^{4} m^2 = 1 + 4 + 9 + 16$. Leaving the upper limit at 5 would add a fifth term and give 55. $\Rightarrow$ **30**

Try it: Rewrite $\sum_{n=3}^{7}(n-2)^3$ with the index starting at 1 and evaluate it.

check: **225**

Skill 2 — Check the ratio before the sum formula

Rule: $\displaystyle\sum_{n=p}^{\infty} ar^{\,n-p} = \frac{a}{1-r}$ **only if** $|r| < 1$; otherwise the series **diverges** and has no sum.

Two things must be named before dividing: r (the constant multiplier between consecutive terms) and a (the term produced by the *stated* lower limit — not automatically the $n = 0$ term).

Example: Evaluate $\sum_{n=0}^{\infty} 6\left(\frac{1}{3}\right)^n$, or state that it diverges.
Step 1. Each term is the previous one times $\frac{1}{3}$, so this is geometric — but the formula is only licensed once the ratio passes the condition.
Step 2. Here $r = \frac{1}{3}$ and $\left|\frac{1}{3}\right| < 1$, so it converges; the lower limit gives $a = 6$, and $\frac{6}{1-\frac{1}{3}} = \frac{6}{\frac{2}{3}}$.
$\Rightarrow$ **9**

Try it: Evaluate $\sum_{n=0}^{\infty} 4\left(\frac{3}{5}\right)^n$, or state that it diverges.

check: **10**

Skill 3 — Reading the limit of a sequence

Rule: for a ratio of polynomials in n, compare *degrees*: divide numerator and denominator by the highest power of n present, then send every $\frac{1}{n^k}$ to 0. Equal degrees leave the ratio of leading coefficients.

Careful: $a_n \to 0$ does **not** make $\sum a_n$ converge — the harmonic series is the standard counterexample. The nth-term test can prove divergence only.

Example: Find $\lim\limits_{n\to\infty} \dfrac{3n+5}{2n-1}$.

Step 1. Numerator and denominator both grow without bound, so comparing degrees is the tool — divide through by the highest power, n.

Step 2. The quotient becomes $\dfrac{3+\frac{5}{n}}{2-\frac{1}{n}}$, and both fractions tend to 0, leaving the ratio of leading coefficients. $\Rightarrow$ $\boldsymbol{\dfrac{3}{2}}$

Try it: Find $\lim\limits_{n\to\infty} \dfrac{n^2+4}{5n^2-n}$.

check: $\boldsymbol{\dfrac{1}{5}}$